Multi-Hazard Monitor: An IoT Safety Station with a Phone Dashboard

LPG, alcohol vapour, temperature, soil saturation and water-level monitoring over a password-free WiFi hotspot

Category	Level	Main board
Internet of Things	High school	ESP32-C3

1. ABSTRACT

The Multi-Hazard Monitor is an Internet-of-Things station that watches five different dangers at once: LPG / cooking gas leaks (MQ-5), alcohol or solvent vapour (MQ-3), high temperature (DHT11), soil that is too dry or so wet that flooding or landslides become likely (soil moisture probe), and rising water or an approaching object (ultrasonic). The ESP32-C3 creates its own open WiFi hotspot; any phone that joins it is taken straight to a live dashboard at 192.168.4.1 that updates every second. A single red LED shows the overall state on the device itself. Sensors that are disconnected are clearly shown as 'not detected' instead of displaying fake values.

2. AIM OF THE PROJECT

- Monitor gas, vapour, temperature, soil moisture and water level with one low-cost board.
- Show live readings on any phone without internet or an app.
- Warn with a simple red LED: off, slow blink, fast blink.
- Be honest about faults: never show a made-up reading when a sensor is missing.

3. PROBLEM STATEMENT AND NEED

Kerala faces LPG accidents at home and landslides and floods during the monsoon. Most monitoring systems watch only one danger and need internet or an app. A single cheap station that watches several dangers and can be read on any phone nearby is useful in homes, farms, schools and hillside settlements.

4. COMPONENTS USED AND COST

Component	Model	Qty	Purpose	Approx. cost
Controller	ESP32-C3 SuperMini	1	Reads sensors, hotspot, web server	Rs. 300
LPG sensor	MQ-5 module	1	Cooking gas / natural gas leaks	Rs. 150
Alcohol sensor	MQ-3 module	1	Alcohol, sanitiser and solvent vapour	Rs. 150
Temperature sensor	DHT11	1	Temperature and humidity	Rs. 110
Soil sensor	Soil moisture probe + module	1	Dry soil / waterlogged soil	Rs. 80
Ultrasonic sensor	HC-SR04	1	Water level rise or object too close	Rs. 90
Indicator	Red 5 mm LED + 220 Ω	1	Overall alarm state	Rs. 5
Resistors	10 kΩ x4, 1 kΩ, 2 kΩ	1 set	MQ output dividers, echo divider	Rs. 15
Power	5 V 2 A USB-C adapter	1	Supply (MQ heaters need 5 V)	Rs. 150
Total				Rs. 1,050

Table 1: Bill of materials (approximate local prices)

5. CIRCUIT CONNECTIONS

Analog sensors must use the ESP32-C3's ADC1 pins (GPIO 0-4) because ADC2 stops working while WiFi is on. The ADC reads about 0-2.5 V reliably, so each 5 V MQ output goes through a 10 kΩ / 10 kΩ divider and the firmware multiplies the reading back by 2.

Part pin	ESP32-C3 pin	Notes
MQ-3 AO	GPIO 0	10k:10k divider
MQ-5 AO	GPIO 1	10k:10k divider
Soil AO	GPIO 3	Module powered from 3.3 V
HC-SR04 TRIG / ECHO	GPIO 4 / 5	ECHO through 1 kΩ / 2 kΩ divider
DHT11 DATA	GPIO 6	10 kΩ pull-up
Red LED	GPIO 9	3.3 V -> 220 Ω -> LED -> GPIO 9 (active LOW)

Table 2: Wiring

6. SYSTEM OVERVIEW

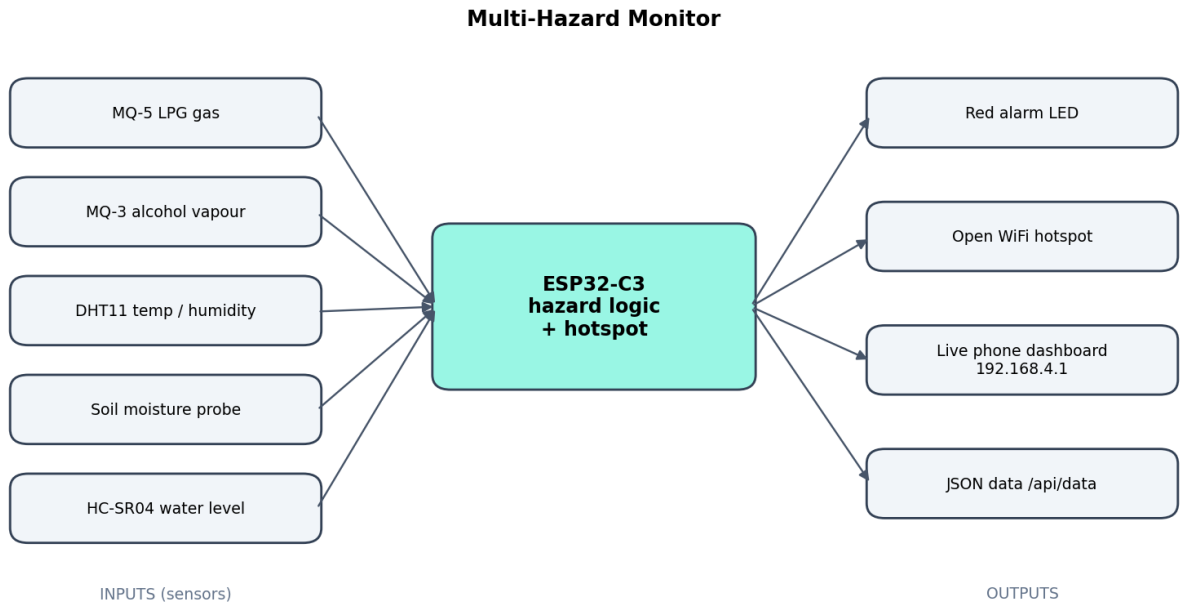


Figure 1: Block diagram

7. HOW I DID THIS (METHOD)

Every analog sensor is read 16 times and averaged with the ESP32's calibrated millivolt function. The soil reading is converted to a percentage using two calibration points (probe in dry air and dipped in water). The DHT11 and ultrasonic sensor are marked missing after 3-4 failed readings in a row, and their cards on the dashboard turn grey. The device runs only as a hotspot (no router), which keeps the WiFi channel fixed so phones never lose the connection. A small DNS server makes phones open the dashboard automatically (captive portal).

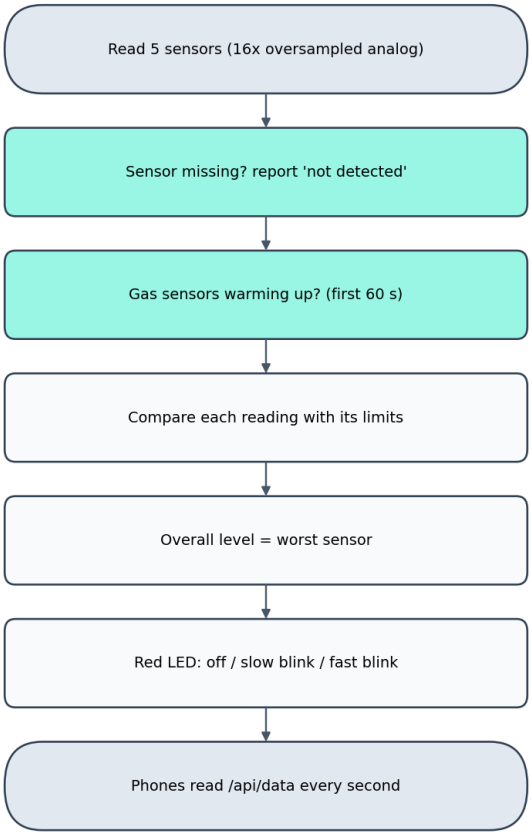


Figure 2: Program flow

8. TESTING AND RESULTS

The firmware was flashed to the real ESP32-C3 and a laptop confirmed the open hotspot (signal -44 dBm, no password). The serial log below was recorded from the board with the gas sensors in clean room air.

Uptime	MQ-3 (mV)	MQ-5 (mV)	Soil raw (mV)
20 s	714	954	350
22 s	712	952	341
24 s	684	930	347

Table 3: Readings logged from the real board

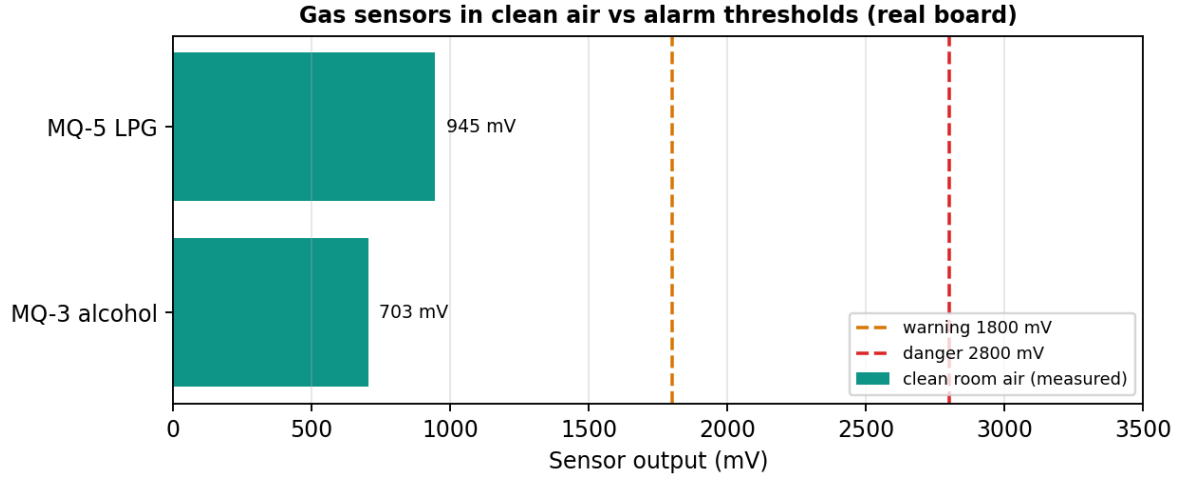


Figure 3: Clean-air gas readings are far below the warning limits

In this first test the soil channel read about 345 mV, which the program treats as fully wet, and the DHT11 and ultrasonic sensor were correctly reported as 'not detected' because they were not yet wired. This showed that the fault reporting works: a missing sensor is never shown as a normal reading.

9. APPLICATIONS

- Kitchens: LPG leak warning.
- Labs and workshops: solvent and alcohol vapour.
- Farms: irrigation reminder when soil is too dry.
- Hill areas: warning when soil is saturated during heavy rain.
- Sumps and drains: rising water alert.

10. ADVANTAGES AND LIMITATIONS

- Five hazards on one board for about a thousand rupees.
- Any phone can view it; no internet or app needed.
- Clear fault reporting.
- Limitation: MQ sensors give relative readings and need calibration in clean air.
- Limitation: an open hotspot can be joined by anyone nearby (it only shows readings).

11. FUTURE IMPROVEMENTS

- Log history to an SD card or the cloud.
- Add a buzzer and SMS alerts.
- Solar power and a waterproof box for outdoor soil monitoring.

12. CONCLUSION

The Multi-Hazard Monitor shows how one small IoT board can watch several everyday dangers at once and share them instantly with every phone nearby, while staying honest about what it can and cannot measure.

13. REFERENCES

- Hanwei Electronics - MQ-3 and MQ-5 datasheets.
- Espressif - ESP32-C3 ADC calibration documentation.
- Arduino-ESP32 WebServer and DNSServer libraries.